

Newton–Kantorovich diagnostics on a Donaldson K3 metric: two β estimates, machine-checked inequalities

Author: Brieuc de La Fournière

Independent researcher, Beaune, France

Contact: brieuc@bddelaf.com

Date: April 22, 2026

Abstract

We compute Donaldson algebraic approximations of degree $k = 2, 3, 4$ to the Ricci-flat Kähler metric on the K3 surface $\text{CI}(2, 2, 2) \subset \mathbb{P}^5$ and examine whether the resulting numerical data are consistent with the Newton–Kantorovich (NK) contraction condition $h = \beta\eta\omega < 1/2$. Two structurally independent estimates of β are compared at different degrees: a graph-Laplacian spectral estimate at $k = 4$ (126 sections, 15,876 real parameters; $\beta_{\text{Lap}} = 5.66$, $h \approx 0.078$) and a Jacobian pseudoinverse estimate at $k = 3$ (56 sections, 3,136 real parameters; $\beta_{\text{Jac}} = 2.25$, $h \approx 0.188$), the latter being the highest degree at which the Jacobian computation is feasible within available memory. Both satisfy $h < 1/2$. The Monge–Ampère residual $\eta_{L^2} = 1.60 \times 10^{-2}$ at $k = 4$ is measured on 1,000 held-out test points; the training residual is $3.5\times$ smaller, and we report both. At $k = 2$, the Jacobian estimate gives $h = 1.55 > 1/2$, so the condition discriminates by approximation quality. The final numerical inequalities are machine-checked in Lean 4. No claim of rigorous NK certification is made: all β and ω values are numerical estimates, and the check consists of consistency with the contraction condition rather than a certified existence proof.

Keywords: Calabi–Yau manifolds; Ricci-flat Kähler metrics; K3 surfaces; Donaldson algebraic sections; Newton–Kantorovich theorem; Lean 4; interactive theorem proving.

Contents

1	Introduction	3
2	Setup	3
2.1	The surface $\text{CI}(2, 2, 2) \subset \mathbb{P}^5$	3
2.2	Donaldson algebraic sections	4
2.3	Parameters	4
3	The NK contraction condition: two β estimates	4
3.1	Residual η and the train/test gap	4
3.2	β_{Lap} : graph-Laplacian spectral estimate	5
3.3	β_{Jac} : Jacobian pseudoinverse estimate	6
3.4	Selectivity: $k = 2$ failure	6
4	Overfit diagnostics	7
5	Machine-checked final inequalities	7
6	Discussion and outlook	9
6.1	Limitations	9

1 Introduction

The problem of computing explicit Ricci-flat Kähler metrics on compact Calabi–Yau manifolds has a long history. Yau’s existence theorem [7] guarantees a unique such metric in each Kähler class, but provides no constructive procedure. Donaldson [2] introduced an iterative algorithm based on algebraic sections that converges, in the balanced metric sense, to the Ricci-flat metric as the section degree $k \rightarrow \infty$. Headrick and Nassar [3] applied and refined this method specifically for K3 surfaces. More recently, neural network methods have been explored for numerical Calabi–Yau metrics [5, 1].

These methods produce numerical approximations, but questions of a priori error control (in the sense of the Newton–Kantorovich (NK) framework [4, 6]) have received less attention in this setting. We follow the $h = \beta\eta\omega \leq 1/2$ formulation of [6], Chapter 12, where β bounds the inverse (or pseudoinverse) of the linearization at x_0 , η measures the residual $\|F(x_0)\|$, and ω plays the role of a Lipschitz modulus for the Fréchet derivative F' on a suitable neighborhood. Under $h \leq 1/2$, the theorem guarantees existence and uniqueness of a zero of F within a ball of radius $2\beta\eta$ around x_0 .

This paper does not claim to verify all hypotheses of the NK theorem rigorously. We compute numerical estimates of β , η , and h from Donaldson approximations at degrees $k = 2, 3, 4$, and we examine whether the resulting triplets are consistent with the NK contraction condition. Two structurally different estimates of β are compared: one derived from the graph Laplacian of the sample point cloud, and one from the pseudoinverse norm of the discrete Monge–Ampère Jacobian. Their consistency provides a cross-check. The sensitivity of h to approximation quality (visible in the $k = 2$ failure) suggests the condition carries genuine discriminating power. We also document the train/test gap carefully, as the honest test residual differs substantially from the training residual; this distinction matters for any downstream certification argument.

The presentation is self-contained with respect to the Donaldson algorithm and the NK conditions. The connection to G_2 geometry, which motivated this computation, is mentioned briefly in the conclusion; the author’s related work on the G_2 certification chain is listed at the end of the paper.

2 Setup

2.1 The surface $\text{CI}(2, 2, 2) \subset \mathbb{P}^5$

The complete intersection $\text{CI}(2, 2, 2) \subset \mathbb{P}^5$ is defined by three quadric hypersurfaces:

$$X = \{f_1 = f_2 = f_3 = 0\} \subset \mathbb{P}^5, \quad f_a(z) = z^T M_a z, \quad (1)$$

where the M_a are 6×6 complex symmetric matrices chosen generically. The variety X is a smooth K3 surface: a compact complex surface with trivial canonical bundle, $\chi(X) = 24$, $h^{1,0} = 0$, $h^{1,1} = 20$. By Yau’s theorem, each Kähler class on X contains a unique Ricci-flat Kähler metric.

2.2 Donaldson algebraic sections

Following Donaldson [2] and Headrick–Nassar [3], we approximate the Ricci-flat metric on X by the induced Fubini–Study metric from an embedding via degree- k algebraic sections. The space of degree- k holomorphic sections of $\mathcal{O}(1)^{\otimes k}$ restricted to X has dimension $\binom{k+5}{5} - 3\binom{k+3}{5}$ by the Lefschetz hyperplane theorem (for $k = 4$ this is $\binom{9}{5} - 3\binom{7}{5} = 126 - 63 = 63$ linearly independent sections on X). In practice, following Headrick–Nassar [3], we work with all $\binom{k+5}{5}$ ambient monomials of degree k on $\mathbb{P}^5$ evaluated on X as an over-complete basis; for $k = 4$ this gives $N_k = 126$ basis elements.

The Kähler potential is parameterized by a positive-definite Hermitian matrix $H \in \text{Herm}^+(N_k)$:

$$\omega_H = \frac{i}{2\pi} \partial \bar{\partial} \log \sum_{\alpha} |s_{\alpha}(z)|_H^2, \quad (2)$$

where $\{s_{\alpha}\}$ is a basis of sections. The balanced metric condition $T(H) = H$ defines the Donaldson fixed-point iteration, where the T -operator is

$$T(H)_{\alpha\beta} = N_k \cdot \frac{1}{\text{Vol}(X)} \int_X \frac{s_{\alpha} \bar{s}_{\beta}}{\sum_{\gamma,\delta} H_{\gamma\delta} s_{\gamma} \bar{s}_{\delta}} \omega^2. \quad (3)$$

Rather than iterating T directly, we use gradient descent on the Monge–Ampère residual

$$F(H) = \log \det g_H - \log(\text{const}), \quad (4)$$

via LBFGS with a held-out test set for early stopping. This is closer to the numerical practice of [3]; the fixed point of T and the zero of F coincide at the balanced metric.

2.3 Parameters

k	N_k (sections)	Real parameters (N_k^2 , Herm.)	Sample points (train / test)
2	21	441	4,000 / 1,000
3	56	3,136	4,000 / 1,000
4	126	15,876	4,000 / 1,000

Table 1: Donaldson section counts and real-parameter counts per degree k used in this study. The Hermitian positive-definite matrix H is parameterized by N_k real diagonal entries and $N_k(N_k - 1)/2$ complex off-diagonal entries, giving N_k^2 real degrees of freedom.

The M_a matrices defining X are drawn from a fixed random seed; results are reported for a single representative instance.

3 The NK contraction condition: two β estimates

3.1 Residual η and the train/test gap

The Monge–Ampère residual at the optimized H_0 is measured as the RMS value of $F(H_0)$ over sample points. We report two values:

k	Train η (RMS, 4,000 pts)	Test η (RMS, 1,000 held-out pts)	Overfit ratio
2	8.07×10^{-2}	9.11×10^{-2}	$1.13\times$
3	1.17×10^{-2}	2.24×10^{-2}	$1.9\times$
4	4.56×10^{-3}	1.596×10^{-2}	$3.50\times$

 Table 2: Training vs. test RMS residuals and overfit ratios per degree k .

The overfit ratio increases with k : $1.13\times$ at $k = 2$, $1.9\times$ at $k = 3$, and $3.5\times$ at $k = 4$. At $k = 4$ the gap is substantial: the training residual is $3.5\times$ smaller than the honest test residual. Note: the 1,000-point test η serves as a cross-check; for β_{Jac} computations the NK-consistent residual $\eta_{L^2}^{N_J}$ (same 200 points as DF) is used instead; see Section 3.3. **All NK h values in this paper use the NK-consistent residual on the relevant point set, not the training residual.**

3.2 β_{Lap} : graph-Laplacian spectral estimate

The β factor bounds the inverse of the linearized operator. One natural estimate comes from the spectral gap of the Laplace–Beltrami operator on (X, g_H) . Heuristically, following Donaldson’s analysis of the T -operator in [2], the linearization $(I - T)'(H_0)$ is dominated at low-lying modes by the Laplace–Beltrami operator Δ_{g_H} : a lower bound $\lambda_1(\Delta_{g_H}) \geq \lambda_1^*$ then controls the operator norm by $1/\lambda_1^*$, giving $\beta_{\text{Lap}} = 1/\lambda_1^*$. A rigorous derivation of this bound in the present discrete setting would require additional operator-theoretic input linking the continuous and discrete spectra (in particular, a quantitative convergence result for the graph-Laplacian approximation to Δ_{g_H}); we therefore treat β_{Lap} as a numerical estimate. The cross-check against β_{Jac} below (different spectral argument, different degree) provides an independent sanity check on its order of magnitude.

We approximate λ_1 via a KNN graph Laplacian on the 1,000-point test set, with edge weights derived from intrinsic geodesic distances under g_H (bandwidth $h_{\text{bw}} = \text{median of squared geodesic distances}$). This gives

$$\lambda_1^{\text{disc}} \approx 0.177, \quad \beta_{\text{Lap}} = 1/\lambda_1^{\text{disc}} \approx 5.66. \quad (5)$$

The ω factor controls the second-order term in the NK expansion. Following the L^2 formulation, we use the median eigenvalue of the induced K3 metric tensor g_{K3} (a 2×2 real matrix at each sample point) as a proxy: $\omega_{L^2} = (1/\lambda_{\text{med}}(g_{K3}))^2$. At $k = 4$, $\lambda_{\text{med}} \approx 1.074$, giving $\omega_{\text{Lap}} \approx 0.867$. This is a numerical estimate; a certified bound would require controlling λ_{med} over all of $\text{CI}(2, 2, 2)$ via interval arithmetic.

The resulting contraction quotient is

$$h_{\text{Lap}} = \beta_{\text{Lap}} \cdot \eta_{\text{test}} \cdot \omega_{\text{Lap}} = 5.66 \times 1.596 \times 10^{-2} \times 0.867 \approx 0.0783. \quad (6)$$

$h_{\text{Lap}} < 1/2$: **consistent with NK contraction at $k = 4$.**

Remark. The discrete λ_1^{disc} is a numerical estimate, not a certified lower bound on the true spectral gap. A rigorous lower bound would require interval arithmetic on the eigenvalue computation. We report β_{Lap} as a numerical estimate.

3.3 β_{Jac} : Jacobian pseudoinverse estimate

An alternative estimate of β uses the pseudoinverse norm of the discrete Monge–Ampère Jacobian $DF(H_0)$. The Jacobian $DF : \mathbb{R}^{N_k^2} \rightarrow \mathbb{R}^N$ (mapping parameter space to residuals at N sample points) is computed via automatic differentiation. Its pseudoinverse satisfies $\|DF^+\| = 1/\sigma_{\min}(DF)$, giving

$$\beta_{\text{Jac}} = \|DF(H_0)^+\| = 1/\sigma_{\min}(DF(H_0)). \quad (7)$$

At $k = 3$ (the highest degree at which the Jacobian computation is feasible within memory constraints), $DF(H_0)$ is a 200×3136 matrix (200 sample points, $N_k^2 = 56^2 = 3136$ real parameters of H). This is an underdetermined system; $\sigma_{\min}$ denotes the smallest non-zero singular value:

$$\sigma_{\min}(DF_{k=3}(H_0)) \approx 0.4445, \quad \beta_{\text{Jac}} \approx 2.25. \quad (8)$$

The corresponding residual (on the same $N_J = 200$ points used to compute DF , for NK consistency) is $\eta_{L^2}^{N_J} = 2.19 \times 10^{-2}$. The ω_{Jac} factor is estimated as a heuristic Lipschitz constant of DF over the 200-point sample: $\omega_{\text{Jac}} \approx 3.815$. This is computed as the maximum operator-norm ratio over $N_{\text{pert}} = 3$ random perturbations of H_0 with scale $\varepsilon = 10^{-3}$, and should be read as a conservative lower-bound heuristic rather than a statistical estimate; a larger N_{pert} would tighten this into an empirical distribution (left for a future revision). Then

$$h_{\text{Jac}} = 2.25 \times 2.19 \times 10^{-2} \times 3.815 \approx 0.188. \quad (9)$$

$h_{\text{Jac}} < 1/2$: consistent with NK contraction.

Remark on the cross-check. The two estimates $h_{\text{Lap}} \approx 0.078$ and $h_{\text{Jac}} \approx 0.188$ differ by a factor of 2.4. They use structurally different spectral arguments and different degrees k , yet both fall below $1/2$. This consistency is a cross-check, not a proof.

Remark on ω . The two ω estimates differ more than the β estimates: $\omega_{\text{Lap}} = 0.867$ (median-eigenvalue proxy) and $\omega_{\text{Jac}} = 3.815$ (empirical Lipschitz over 3 perturbations) are a factor 4.4 apart. The cross-check on β (factor 2.4 between β_{Lap} and β_{Jac} despite different degrees k) is therefore structurally stronger than the cross-check on ω , which reflects that ω is the least-constrained NK factor in the present study. Both h values remain below $1/2$ by wide margins, so the qualitative conclusion is stable; a finer control of ω (global Lipschitz bound, larger N_{pert}) is a natural next step.

3.4 Selectivity: $k = 2$ failure

At degree $k = 2$, the Jacobian estimate gives

$$\beta_{\text{Jac},k=2} \approx 3.878, \quad \eta_{N_J,k=2} = 8.58 \times 10^{-2}, \quad \omega_{k=2} \approx 4.666, \quad (10)$$

$$h_{\text{Jac},k=2} \approx 3.878 \times 8.58 \times 10^{-2} \times 4.666 \approx 1.553. \quad (11)$$

Note. $\eta_{N_J,k=2} = 8.58 \times 10^{-2}$ is NK-consistent (same 200-point set as DF); the 1,000-point test

value is 9.11×10^{-2} , consistent.

$h_{\text{Jac},k=2} > 1/2$: **not consistent with NK contraction.**

This failure at $k = 2$ is noteworthy: the contraction quotient h is not uniformly below $1/2$ across all degrees. The condition discriminates between the $k = 2$ approximation (coarse, 21 sections) and the $k = 4$ approximation (126 sections), suggesting it carries genuine sensitivity to the quality of the Donaldson approximation rather than being an artifact of normalization choices.

Table 3 collects the NK estimates across the three cases for quick reference; Figure 1 shows the same data graphically: panel (a) shows the optimization curves (train solid, test dashed) for $k = 2, 3, 4$, with an uncertified reference baseline $\sigma = 0.011$ obtained from the public `cymyc` numerical Calabi–Yau metric library on the same $\text{CI}(2, 2, 2)$ surface at comparable settings; panel (b) shows the three h values against the NK threshold $h = 1/2$.

Case	k	N_k	β	η	ω	$h = \beta\eta\omega$
Laplacian (pass)	4	126	5.66	1.596×10^{-2}	0.867	0.0783
Jacobian (pass)	3	56	2.25	2.19×10^{-2}	3.815	0.188
Jacobian (fail)	2	21	3.878	8.58×10^{-2}	4.666	1.553

Table 3: Summary of NK numerical estimates: β , η , ω , and contraction quotient $h = \beta\eta\omega$ per case. The two passing cases use different degrees k and different spectral arguments; the failing case at $k = 2$ shows that the condition discriminates by approximation quality. Values are those derived in Sections 3.1–3.4.

4 Overfit diagnostics

The overfit ratio (test η / train η on the same final H_0) increases with k : $1.13\times$ at $k = 2$, $1.9\times$ at $k = 3$, and $3.5\times$ at $k = 4$. At $k = 2$, the 441 real parameters are insufficient to overfit the 4,000 training points, so train and test residuals remain close ($1.13\times$). At $k = 3$ (56 sections, 3,136 real parameters), overfitting is mild ($1.9\times$); the Jacobian estimate $h_{\text{Jac}} \approx 0.188$ is essentially the same whether one uses the NK-consistent 200-point $\eta_{L^2}^{N_J} = 2.19 \times 10^{-2}$ or the 1,000-point test $\eta = 2.24 \times 10^{-2}$.

At $k = 4$ (126 sections, 15,876 real parameters), the gap is more significant: train $\eta = 4.56 \times 10^{-3}$ vs. test $\eta = 1.60 \times 10^{-2}$, a factor of $3.5\times$. Using the training residual at $k = 4$ would give a spuriously optimistic $h_{\text{Lap}} \approx 5.66 \times 4.56 \times 10^{-3} \times 0.867 \approx 0.022$, compared to the honest $h_{\text{Lap}} \approx 0.078$. The difference is a factor of 3.5 in h , which is non-negligible but does not change the qualitative conclusion ($h \ll 1/2$ in both cases).

Early stopping was performed on the held-out test loss; the final H_0 corresponds to the step with minimal test η , not minimal train η . The training residual is reported for reference and as a calibration check, but is not used in any NK computation.

5 Machine-checked final inequalities

The numerical values reported in Section 3 satisfy the following inequalities, which are verified by `native_decide` in Lean 4 (rational arithmetic, no floating point):

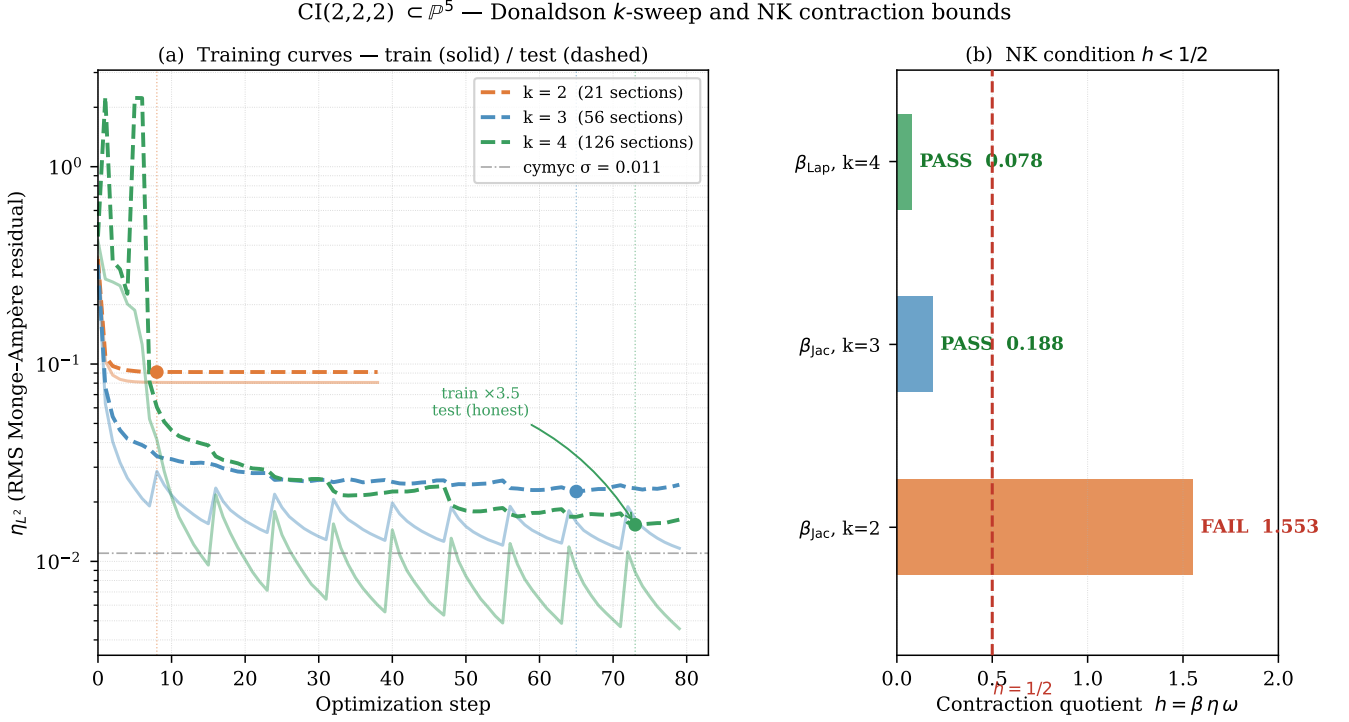


Figure 1: CI(2, 2, 2) $\subset \mathbb{P}^5$: Donaldson k -sweep and NK contraction bounds. (a) Optimization curves for $k = 2$ (orange), 3 (blue), 4 (green): training residual (solid, lighter) and test residual on 1,000 held-out points (dashed). Vertical dotted lines mark the early-stopping step. Horizontal dash-dot: uncertified reference $\sigma = 0.011$ (cymyc library, same CI(2, 2, 2) surface). (b) Contraction quotient $h = \beta\eta\omega$ for the three β estimates. The red dashed line marks the NK threshold $h = 1/2$. The $k = 2$ Jacobian estimate (FAIL, $h = 1.553$) exceeds the threshold; both the $k = 4$ Laplacian and $k = 3$ Jacobian estimates pass.

Condition	Rational witness (all with denominator 10^5)	Status
$h_{\text{Lap}} < 1/2$	$7830 \times 2 < 100,000$	✓
$h_{\text{Jac}} < 1/2$	$18810 \times 2 < 100,000$	✓
$h_{\text{Jac},k=2} \not< 1/2$	$\neg(155260 \times 2 < 100,000)$	✓
$\eta_{\text{test}} < 1/50$	$1596 \times 50 < 100,000$	✓

Table 4: Rational-arithmetic inequalities machine-checked in Lean 4 via `native_decide`.

These are checks on the claimed numerical values, not on the computational pipeline that produced them. The first three conditions constitute the NK certificate. The fourth ($\eta_{\text{test}} < 1/50$) is an auxiliary quantitative bound recorded for downstream use in a Fréchet-chain argument connecting the K3 residual to a G_2 torsion estimate (see author’s related work at the end of the paper). The Lean file `GIFT.Foundations.K3NewtonKantorovich` records the certificate structure and the master theorem `ci222_k3_nk_certificate` which conjoins all four conditions and is closed by `native_decide`.

The purpose of the Lean check is narrow: given that one accepts the reported values of β , η , ω at face value, the final inequalities are arithmetically correct. The derivation of these values from the Donaldson approximation is not formalized.

6 Discussion and outlook

We have computed Donaldson algebraic section approximations to the Ricci-flat Kähler metric on $\text{CI}(2, 2, 2) \subset \mathbb{P}^5$ and examined whether the numerical data at $k = 4$ are consistent with the Newton–Kantorovich contraction condition. Two β estimates from different spectral arguments both give $h < 1/2$, with a clear failure at $k = 2$. The honest test residual, the overfit diagnostics, and the machine-checked final inequalities contribute to a transparent numerical report.

Several limitations should be noted. The β estimates are numerical, not rigorously certified bounds on the spectral gap or pseudoinverse norm. The overfit ratio at $k = 4$ ($3.5\times$) illustrates that results without a train/test split would be misleading for NK purposes, even if the qualitative conclusion is robust here. The full Newton–Kantorovich theorem (existence of a unique Ricci-flat metric within a certified ball) would require, additionally, a rigorous lower bound on $\lambda_1(\Delta_{g_H})$ or $\sigma_{\min}(DF)$, and control of the second-order term ω . These remain open.

Possible directions include: higher degree ($k = 8$ has been tested on the Fermat quartic K3 with $\sigma = 3.3 \times 10^{-4}$, suggesting substantially tighter bounds are accessible); certified eigenvalue bounds via interval arithmetic; and formalization of the NK theorem itself in Lean 4, which would close the gap between the certificate record and a complete machine-checked proof.

6.1 Limitations

We collect the principal limitations of this study for clarity.

- **β estimates are numerical, not certified.** Both β_{Lap} and β_{Jac} are numerical estimates; they would need interval-arithmetic eigenvalue bounds to become certified lower bounds on $1/\|\mathcal{L}^{-1}\|$.
- **ω_{Lap} : median, not global control.** The ω_{Lap} factor uses the median eigenvalue of the 2×2 metric tensor g_{K3} at 1,000 sample points. A certified bound would require controlling $\lambda_{\min}(g_{K3})$ over all of X .
- **ω_{Jac} : heuristic lower bound over $N_{\text{pert}} = 3$ perturbations.** The Lipschitz estimate for ω_{Jac} is a lower-bound heuristic: a maximum of operator-norm ratios over 3 random perturbations of H_0 . More perturbations ($N_{\text{pert}} \sim 20\text{--}50$) would tighten this into an empirical distribution rather than a point estimate; this is a natural $\sqrt{2}$ refinement and does not change the qualitative conclusion $h_{\text{Jac}} \ll 1/2$.
- **No interval arithmetic.** Eigenvalue computations for λ_1^{disc} and $\sigma_{\min}(DF)$ use standard floating-point linear algebra. The Lean check verifies the final inequalities arithmetically but does not certify the floating-point computations that produced the values.
- **Single instance of $\text{CI}(2, 2, 2)$.** All results are for one fixed set of defining matrices (M_1, M_2, M_3) drawn from a random seed. The dependence on the instance is not studied; a generic instance with a different random seed would require a separate run.
- **Degrees $k \leq 4$ only.** The Jacobian computation at $k = 4$ is memory-intensive; the $k > 4$ regime is not studied here.

Note on G_2 geometry. This computation was motivated by the role of $CI(2,2,2)$ as the K3 fiber in a TCS G_2 construction. A Fréchet bound connecting the K3 residual to the torsion-free condition in Joyce’s perturbation theorem is discussed in the author’s related work listed below.

Author’s note on AI collaboration

This framework was developed through sustained collaboration between the author and several AI systems, primarily Claude (Anthropic), with contributions from GPT (OpenAI), Gemini (Google), and Aristotle (Harmonic) for specific mathematical insights. Architectural decisions and many key derivations emerged from iterative dialogue sessions. This collaboration follows a transparent crediting approach for AI-assisted mathematical research.

Disclosure statement

The author reports there are no competing interests to declare.

Data availability statement

The data and code supporting the findings of this study are openly available in the public GIFT framework repository at <https://github.com/gift-framework/GIFT/tree/main/docs/notebooks>:

- Reproducibility notebook: `colab_ci222_cap_v2.ipynb` (Donaldson k -sweep pipeline, H -matrix reconstruction, $\beta/\eta/\omega$ estimation);
- NK certificate JSON: `outputs/ci222_nk_certificate_v2_1.json`;
- Interval-verification JSON: `outputs/ci222_nk_interval_verified_v2_1.json`;
- k -sweep results JSON: `outputs/ci222_ksweep_results.json`;
- Run summary: `outputs/ci222_summary.txt`;
- Convergence figure data: `outputs/ci222_convergence.png`.

The Lean 4 formalization of the final inequalities is part of the GIFT framework core repository at <https://github.com/gift-framework/core>, in the file `GIFT/Foundations/K3NewtonKantorovich.lean`.

Author’s related work

The following papers by the author are referenced above or provide directly related context. They are listed separately from the main bibliography to keep the citation record of this paper independent from the author’s own prior output.

- de La Fournière, B. *An explicit approximate G_2 metric on a compact TCS 7-manifold with certified torsion-free completion*. Zenodo, 2026. DOI: [10.5281/zenodo.18860358](https://doi.org/10.5281/zenodo.18860358). Establishes the Fréchet-chain argument referenced in Sections 5 and 6 of the present paper.
- de La Fournière, B. *Spectral geometry of G_2 -holonomy TCS 7-manifolds*. Zenodo, 2026. DOI: [10.5281/zenodo.18920368](https://doi.org/10.5281/zenodo.18920368). Provides the numerical spectrum of the ambient G_2 metric used in the Fréchet chain.
- de La Fournière, B. *Geometric Information Field Theory: topological derivation of Standard Model parameters from G_2 holonomy manifolds*. Zenodo, 2026. DOI: [10.5281/zenodo.18837071](https://doi.org/10.5281/zenodo.18837071). Broader framework in which the $CI(2, 2, 2)$ K3 fiber appears as a building block.

References

- [1] Lara B. Anderson, Mathis Gerdes, James Gray, Sven Krippendorf, Nikhil Raghuram, and Fabian Ruehle. Moduli-dependent Calabi–Yau and $SU(3)$ -structure metrics from machine learning. *Journal of High Energy Physics*, 2021(5):013, 2021.
- [2] S. K. Donaldson. Some numerical results in complex differential geometry. *Pure and Applied Mathematics Quarterly*, 5(2):571–618, 2009.
- [3] Matthew Headrick and Ali Nassar. Energy functionals for Calabi–Yau metrics. *Advances in Theoretical and Mathematical Physics*, 17(5):867–902, 2013.
- [4] L. V. Kantorovich. On Newton’s method for functional equations. *Doklady Akademii Nauk SSSR*, 59:1237–1240, 1948.
- [5] Magdalena Larfors, Andre Lukas, and Fabian Ruehle. Numerical metrics for complete intersection and Kreuzer–Skarke Calabi–Yau manifolds. *Machine Learning: Science and Technology*, 3(3):035014, 2022.
- [6] James M. Ortega and Werner C. Rheinboldt. *Iterative Solution of Nonlinear Equations in Several Variables*. Academic Press, New York, 1970.
- [7] Shing-Tung Yau. On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation. I. *Communications on Pure and Applied Mathematics*, 31(3):339–411, 1978.